

Maru Sathvik Reddy

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SKILLS

- Languages:** Python, C++
 - AI/ML Libraries:** PyTorch, Transformers, Scikit-learn, NumPy, Pandas, OpenCV, LangChain, LangGraph
 - Tools/Platforms:** MySQL, Git, GitHub, VS Code, Docker, Redis
 - Soft Skills:** Problem-Solving, Teamwork, Adaptability

PROJECTS

- Multi-Agent Research Assistant:** | [GitHub](#)

Mar' 26

 - Engineered a fully local multi-agent pipeline in LangGraph with 5 specialized agents (Supervisor, Planner, Researcher, Analyst, Writer) communicating via a stateful routing loop with async parallel scraping and in-flight lock to prevent duplicate executions.
 - Implemented a 3-layer Redis Stack cache — prompt-level LLM cache, versioned report storage, and HNSW vector search via all-MiniLM-L6-v2 — enabling hybrid reports that merge cached context with fresh scraped data.
 - Developed an agentic chat layer with intent-based tool routing, session-persisted context, and crash recovery via MemorySaver checkpointing and SHA256-derived thread IDs.

Tech: Python, LangGraph, LangChain, Ollama, FastAPI, Redis, Docker, Nginx, Streamlit
- Multi-Modal Retrieval Augmented Generation:** | [GitHub](#)

Mar' 26

 - Built a RAG pipeline supporting documents, images, and timestamp-aware video ingestion using LLaMA3 + quantized LLaVA for context-grounded generation.
 - Designed a FastAPI backend with semantic chunking, batched GPU embeddings, ChromaDB vector indexing, and streaming token-level responses.
 - Reduced latency and hallucination via model quantization, frame-differencing, and efficient vector retrieval strategies.

Tech: Python, PyTorch, Transformers, LangChain, FastAPI, ChromaDB, Sentence-Transformers, OpenCV, Ollama, Streamlit, Docker
- Autoregressive Transformer for Python Code Generation:** | [GitHub](#)

Nov' 25

 - Trained a decoder-only Transformer for autoregressive Python code generation on a subset of the CodeParrot dataset with GPT-style tokenization.
 - Architected causal self-attention, positional and token embeddings, and a next-token prediction head entirely from scratch in PyTorch.
 - Established a training and inference pipeline supporting prompt-based generation with greedy decoding.

Tech: Python, PyTorch, Transformer Decoder, Self-Attention, GPT Tokenization, NLP

TRAINING

- Machine Learning - Elevate Labs**

Apr' 25 – May' 25

 - Gained hands-on experience in Supervised and Unsupervised Machine Learning, with a strong understanding of core algorithms, feature engineering, and model evaluation techniques.
 - Built a resume ranking system using cosine similarity to match resumes with job descriptions, automating candidate screening through text preprocessing and relevance scoring.

Tech: Python, Pandas, Scikit-learn, NLP

CERTIFICATES

- Generative AI, LLM & RAG | [GeeksforGeeks](#) Feb' 26
 - Foundational ML concepts | [AWS ML Exam Basics](#) Oct' 25
 - A Guide to Machine Learning with Data Science | [Cipher Schools](#) Jul' 25
 - Deep Learning with TensorFlow | [IBM Cognitive Class](#) Mar' 25

ACHIEVEMENT

- Achieved Global Rank 588 (Top 15%) in Kaggle's "Predicting Road Accident Risk" competition Nov' 25

EDUCATION

- Lovely Professional University**

Punjab, India
Aug' 23 - Present

Bachelor of Technology - Computer Science and Engineering; CGPA: 7.28
- Sri Chaitanya Junior College**

Hyderabad, Telangana
Jul' 21 - Mar' 23

Intermediate; Percentage: 88.9
- Wisdom Spaes School**

Metpally, Telangana
Jul' 20 - Mar' 21

Matriculation; Percentage: 100